Sidra Tul Muntaha

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Portfolio | sidra-tul-muntaha-ghouri | Sidra-Tul-Muntaha-Ghouri

Personal Statement

Aspiring researcher and Electrical Engineering graduate with a strong inclination toward smart grids, intelligent energy systems, and data-driven analysis, seeking a Graduate/Teaching Assistant role. Passionate about assisting in teaching, supervising labs, and contributing to research that advances sustainable and efficient power systems.

Education

NED University of Engineering and Technology, Bachelors of Engineering-Electrical Engineering — Dec. 2016 to Jan. 2020

- **GPA:** 3.78/4.0 (Transcript)
- **Thesis:** Modeling of Axial Flux Motor using Comsol Multiphysics for Wind Turbine (Report)
- **Coursework:** Alternate Energy, Signal Processing, Feedback Control System, Electrical Power Generation - Distribution and Protection, Computer Programming, Data Structures & Algorithm, Numerical Methods, Differential Equations, and Probability & Statistics

Skills

Languages: Python, SQL, C++

Software: MATLAB, NI Multisim, Arduino IDE, AutoCAD, Google Colab, Visual Studio, Simaris, MS Excel, DiaLUX, Etap, Eplan Electric P8

ML Frameworks/Libraries: Pandas, Numpy, Scikit-learn, Matplotlib, Searborn, Tensorflow, Streamlit

Professional Experience

ACDC Electrical Services, Senior Design Engineer & Team Lead — Karachi, Pakistan, Jun. 2024 to Present

- Lead a team of 4 Engineers, Manage task and work flow to achieve project objectives and ensure quality deliverables in the set deadlines.
- Design Switchgear panels tailored to client specifications using Eplan and AutoCAD.
- Perform load flow analysis using ETAP to ensure optimal system performance.
- Coordinate and collaborate with 2 international clients, conducting meetings to discuss design proposals, project timelines, and technical clarifications.

Sunbeam Engineers (Pvt.) Ltd., Design Engineer — Karachi, Pakistan, May 2023 to May 2024, 1 Year

- Design Control panels, Distribution Board, Motor Control Center with starters such as direct online, star-delta, soft starter, VFD, Automatic Power Factor Correction, ATS/AMF Panel, Generator Synchronization Panel.
- Calculate the load, circuit breaker, contactors, cables and the bus-bar size required for LV equipment.
- Utilize EPLAN and AutoCAD to create precise schematic drawings, detailing all electrical components, wiring, and connections for switchgear panels.
- Collaborate with cross-functional teams, including electrical engineers, technicians, project managers, and sales teams, to ensure successful project execution and customer satisfaction.

The Spark Foundation, Data Science & Business Analytics Intern.

Remote
Feb. 2023 to Mar. 2023
2 months

- Worked on 3 real-world projects(), showcasing expertise in data manipulation, exploratory data analysis (EDA), and data visualization.
- Employed a diverse toolset, including Microsoft Excel, SQL, Microsoft Power BI, and Python, to extract insights from datasets exceeding 36,000 instances.

  Overall, my virtual internship in Data Science & Business Analytics provided me with practical experience in data quality analysis, creating analytical dashboards, performing data analytics, and delivering data-driven presentations.

Accrescent Engineers (Pvt.) Ltd., Design & Estimation Engineer.

Karachi, Pakistan
Dec. 2021 to May 2023
1 year 6 months

- Estimated more than 50 industrial-scale Building Management System (BMS) control projects.
- Improved costing sheets using advanced Excel functions, leading to a 25% increase in accuracy and a 40% reduction in time consumption.
- Conducted thorough analysis of all client-provided documents, drawings, specifications, and tender materials to ensure accurate project estimation and planning.
- Prepared BMS control and network diagrams using AutoCAD, Microsoft Visio, and CorelDRAW.
- Ensured compliance with project standards and client requirements by preparing comprehensive documentation under the direction of the Lead Manager for approval and execution.

Achievements

- Received a laptop under the Prime Minister's initiative for exceptional academic performance and merit.
- Awarded a merit-based scholarship for three consecutive years during Bachelor's studies.

Projects

Dr. AI, Demo Video()

- Email Extractor is an API based Data Science project, deployed on Streamlit interface. The main objective of this project is to extract emails from a Gmail account and save them as a .csv file to be used for analysis and records
- Used - **Python & IMAP API**.

Yelp Data Analysis, Using SQL()

- This project involved a thorough business data analysis, employing advanced SQL techniques to extract meaningful insights.
- The workflow encompassed data exploration, inferences, and an exploration of popular business categories, presenting findings systematically with code snippets and resulting data tables.
- Used - **SQL**.

Courses And Certificates

***IoT Devices*,** University of Illinois Urbana-Champaign ( )

SQL for Data Science, University of California, Davis ( )

Time Series Analysis in Python, DataCamp( )

Machine Learning With Python, IBM ( )

Power BI, 365 Financial Analyst ( )

Introduction to Data Analytics, IBM ( )

Teamwork Skills: Communicating Effectively in Groups, University of Colorado Boulder ( )

Volunteer Work

Section Leader, Stanford's Code in Place 2024

Remote
Apr. 2024 to May. 2024
1 month

- Code In Place is an online program based on Stanford University's Course "Introduction to Python - CS106A. And as a Section Leader, I taught Python with Karel the Robot, Console Programming, and Graphics Programming to international students, fostering a collaborative and interactive learning environment.

  (Teaching Demo Video)

Omdena, Machine Learning Engineer

Remote
Oct. 2023 to Dec. 2023
3 months

- Collaborated with the Exploratory Data Analysis (EDA) team to extract valuable insights from the data, gained practical insights into assessing data quality and ensuring data accuracy. This involved identifying and resolving issues related to data completeness, consistency, and integrity.
- Worked with deployment team to deploy the designed model using Streamlit framework.

Omdena, Project Collaborator.

Remote
Jan. 2023 to Mar. 2023
3 months

Project: Advancing Weather Prediction using Machine Learning & Python.

- Collaborated with Senior Data Scientist to gather data from Openweathermap API.
- Cleaned and analyzed data using Pandas and Numpy libraries, and visualized using Seaborn and Matplotlib libraries.
- Conducted exploratory data analysis to identify key insights of the data.
- Participated in team meetings to discuss project progress, challenges, and potential solutions. (Notebook Link)